Text as Pieces

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected exact-tokenization record.

Apply the contract, encode/decode, compare sequence lengths and audit preprocessing claims.

Keep one habit: Keep source text, vocabulary/rule version, ordinary/special counts and partial failures visible.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Feedback Loops

Nearest Neighbor Reasoning

Sampling and Representation

W1 · Apply longest matching and case sensitivity

> IDs identify vocabulary entries, not meaning.

Your evidence record

- Original case-sensitive vocabulary:0=end(special),1=cat,2=c,3=a,4=t,5=s,6=one literal space,7=sat. Longest ordinary prefix wins; preserve case/punctuation/space. Unsupported characters stop encoding with a partial trace and zero-based character position. Decode ordinary text IDs by concatenating their strings; state separately if an end marker is appended. Do not reuse the separate book/books vocabulary.

- Explain why sat wins over s at the start of sat and why uppercase C is unsupported.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Encode, decode and declare the count

> Keep four text tokens separate from an appended end marker.

Your evidence record

- Original case-sensitive vocabulary:0=end(special),1=cat,2=c,3=a,4=t,5=s,6=one literal space,7=sat. Longest ordinary prefix wins; preserve case/punctuation/space. Unsupported characters stop encoding with a partial trace and zero-based character position. Decode ordinary text IDs by concatenating their strings; state separately if an end marker is appended. Do not reuse the separate book/books vocabulary.

- Encode cats sat using the given **vocabulary**, decode it, and count with and without an appended end marker.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Compare character and chunk vocabularies

> The source stays the same when representation changes.

Your evidence record

- Original case-sensitive vocabulary:0=end(special),1=cat,2=c,3=a,4=t,5=s,6=one literal space,7=sat. Longest ordinary prefix wins; preserve case/punctuation/space. Unsupported characters stop encoding with a partial trace and zero-based character position. Decode ordinary text IDs by concatenating their strings; state separately if an end marker is appended. Do not reuse the separate book/books vocabulary.

- Encode cats sat with character **token**s only. Explain the difference from four text **token**s in the original **vocabulary**.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Audit undeclared preprocessing

> An altered source cannot be called an exact original round trip.

Your evidence record

- Original case-sensitive vocabulary:0=end(special),1=cat,2=c,3=a,4=t,5=s,6=one literal space,7=sat. Longest ordinary prefix wins; preserve case/punctuation/space. Unsupported characters stop encoding with a partial trace and zero-based character position. Decode ordinary text IDs by concatenating their strings; state separately if an end marker is appended. Do not reuse the separate book/books vocabulary.

- An encoder lowercases Cat and removes punctuation before **encoding**, but labels its **output** an exact **encoding** of the original. Explain what must be recorded or changed.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Trace a new input independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Original case-sensitive vocabulary:0=end(special),1=cat,2=c,3=a,4=t,5=s,6=one literal space,7=sat. Longest ordinary prefix wins; preserve case/punctuation/space. Unsupported characters stop encoding with a partial trace and zero-based character position. Decode ordinary text IDs by concatenating their strings; state separately if an end marker is appended. Do not reuse the separate book/books vocabulary.

Under the original **vocabulary** encode sat cats, decode it and count text **token**s. Then trace sat cat! and state the exact failure position using zero-based character positions.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Trace a new input independently

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Tokenize book descriptions before any text-based matching, while preserving the original source text.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Encode and decode for **Token**isation and **vocabulary**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Token**isation and **vocabulary**. Proposed explanation: “A **token** always equals a word”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Toy **token**izer A splits 'unhelpful' into un/help/ful; B keeps it whole. Count **token**s for each and explain why a three-word sentence need not use three **token**s.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Toy **token**izer A splits rainbows as [rain][bow][s]; B uses [rainbows].

Count **token**s and state the limit of the example.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A classroom **token**izer splits sunflower into sun and flower. How many pieces result? Must every real **token**izer split it the same way?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 25 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Toy **token**izer A splits 'unhelpful' into un/help/ful; B keeps it whole. Count **token**s for each and explain why a three-word sentence need not use three **token**s.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Under the original **vocabulary** encode sat cats, decode it and count text **token**s. Then trace sat cat! and state the exact failure position using zero-based character positions.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

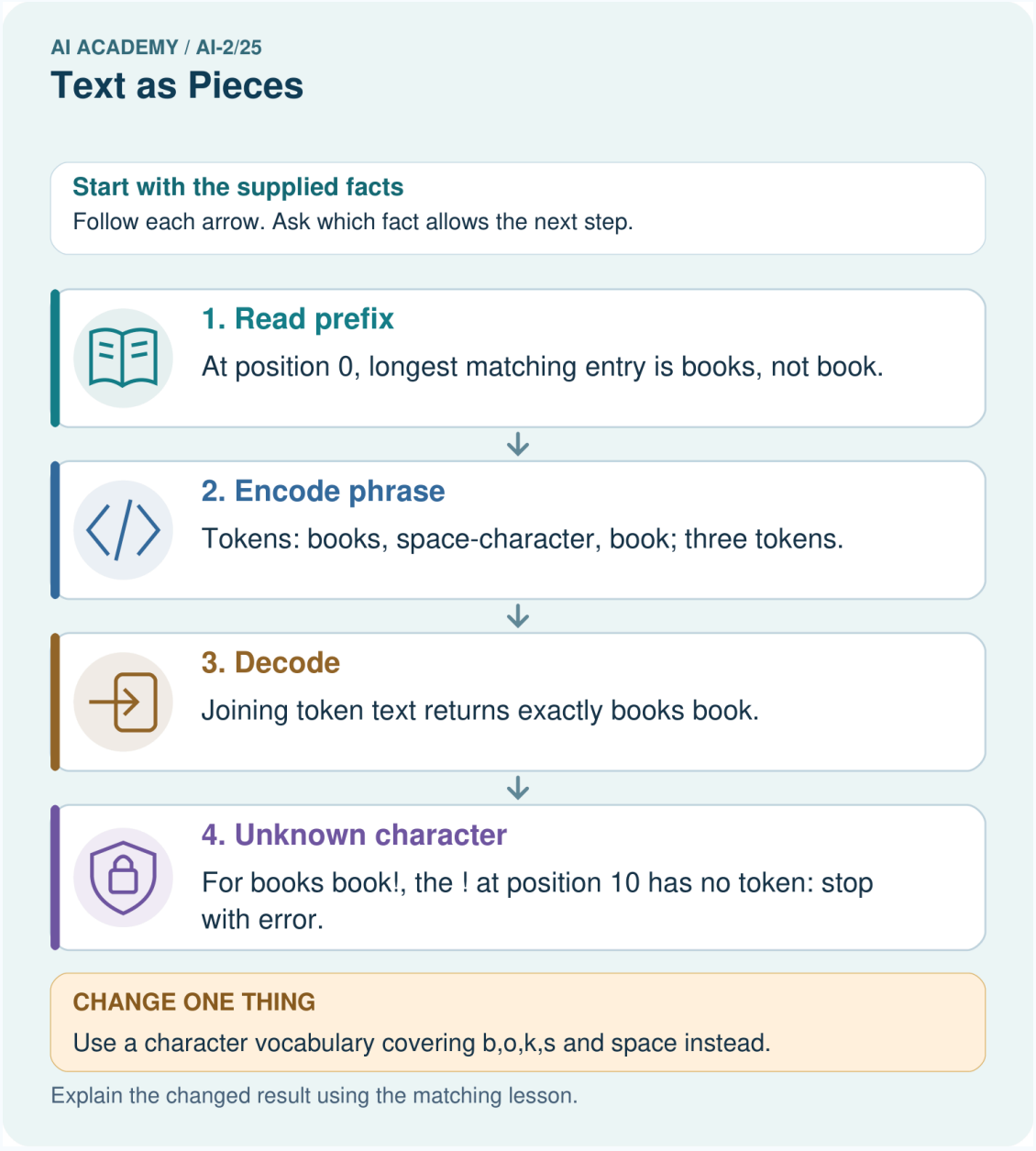
> Keep the explanation and the support you used.

Use this when helpful

A fictional **token**izer converts a short catalogue phrase into **token**s.

Vocabulary entries: book, books and the space character. At each position choose the longest matching entry; preserve the actual space.

Text is “books book”. Exclamation marks have no **token**; zero-based character positions are used.



Follow the labelled picture with the matching explanation. Open the vector diagram

Why is the unknown ! at position 10 rather than position 9?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Document the recommender's text representation and unknown-**token** behavior rather than assuming every model counts words identically.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Document the recommender's text representation and unknown-**token** behavior rather than assuming every model counts words identically.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional **token**izer converts a short catalogue phrase into **token**s.
- Which labelled step matches this detail? For books book!, the ! at position 10 has no **token**: stop with error.
- 1. Unknown character
- 2. Decode

- 3. Encode phrase
- 4. Read prefix
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Unknown character For books book!, the ! at position 10 has no **token**: stop with error.

A **token** is defined by the **vocabulary** and **encoding** rule; it need not be one word or one character. Longest matching matters where **vocabulary** entries overlap. Counting text **token**s here does not count letters.

Preserve spaces during **decoding** so the reconstructed text matches the original exactly.

Changed-case question

- Changed-case practice: Use a character **vocabulary** covering b,o,k,s and space instead. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The same phrase uses ten character **token**s.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The same phrase uses ten character **token**s. There are five letters, one space and four letters; its decoded text remains unchanged.

Repair and save your work

Review Feedback Loops (ai-2/24). Goal: Implement a deterministic recommender. Short practice: What extra field would reveal why comparing raw clicks is misleading?. Return to this week and explain what you changed.

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why do k=1 and k=3 disagree for the original query?. Return to this week and explain what you changed.

Review Sampling and Representation (ai-2/19). Goal: Define a question and target population. Short practice: Can **80%** response be reported as **80%** of the whole grade responding?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

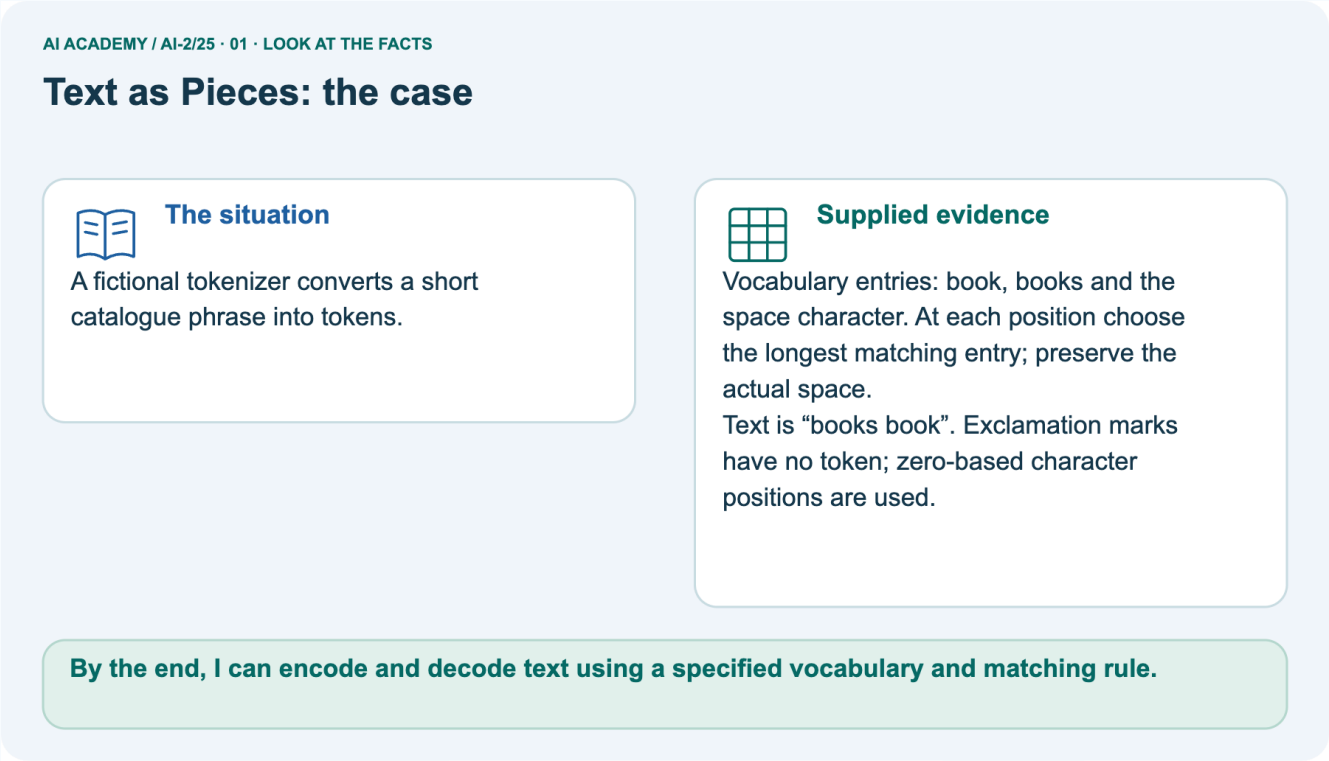
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Text as Pieces: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

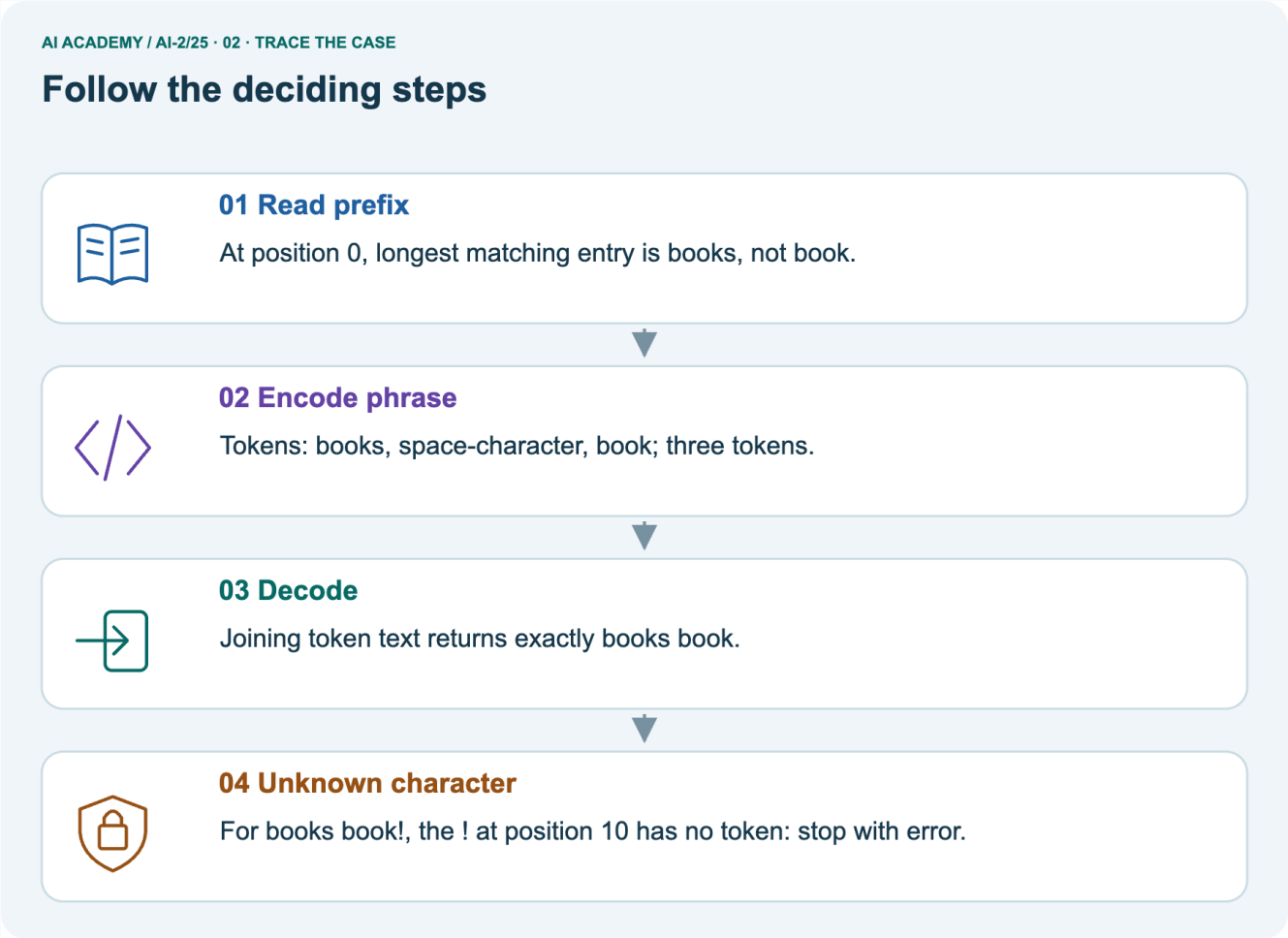
A fictional **token**izer converts a short catalogue phrase into **token**s. Supplied facts: **Vocabulary** entries: book, books and the space character. At each position choose the longest matching entry; preserve the actual space.

Text is “books book”. Exclamation marks have no **token**; zero-based character positions are used.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



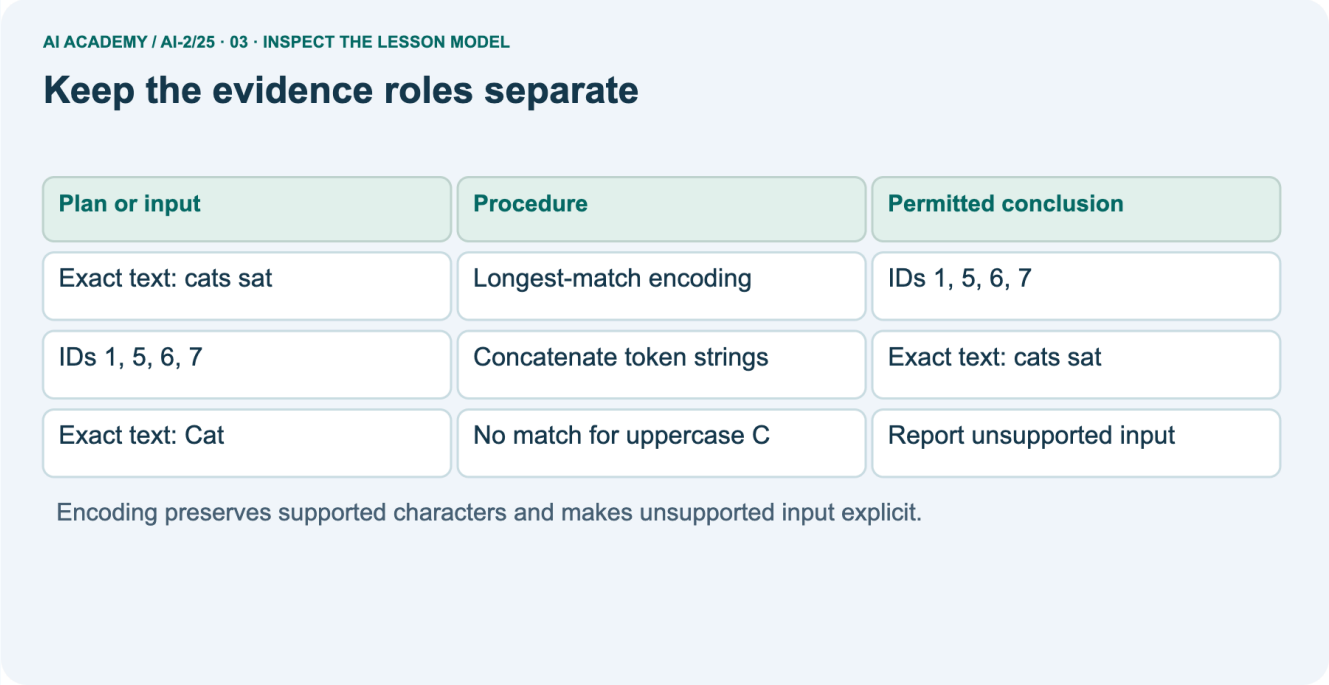
Follow the labelled picture with the matching explanation. Open the vector diagram

Read prefix: At position 0, longest matching entry is books, not book. → Encode phrase: **Token**s: books, space-character, book; three **token**s. → Decode: Joining **token** text returns exactly books book. → Unknown character: For books book!, the ! at position 10 has no **token**: stop with error.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep the evidence roles separate



Follow the labelled picture with the matching explanation. Open the vector diagram

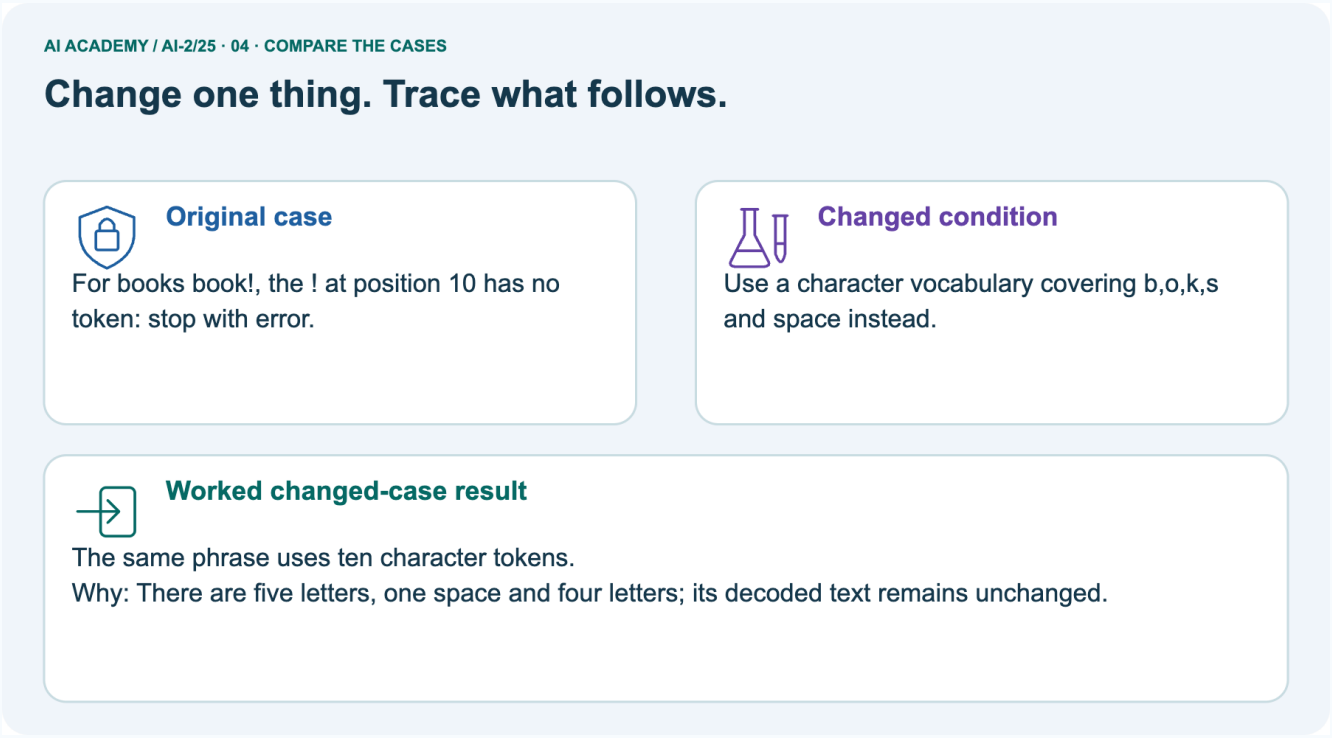
Keep the **evidence** roles separate. Exact text: cats sat | Longest-match **encoding** | IDs 1, 5, 6, 7. IDs 1, 5, 6, 7 | Concatenate **token** strings | Exact text: cats sat.

Exact text: Cat | No match for uppercase C | Report unsupported **input**

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

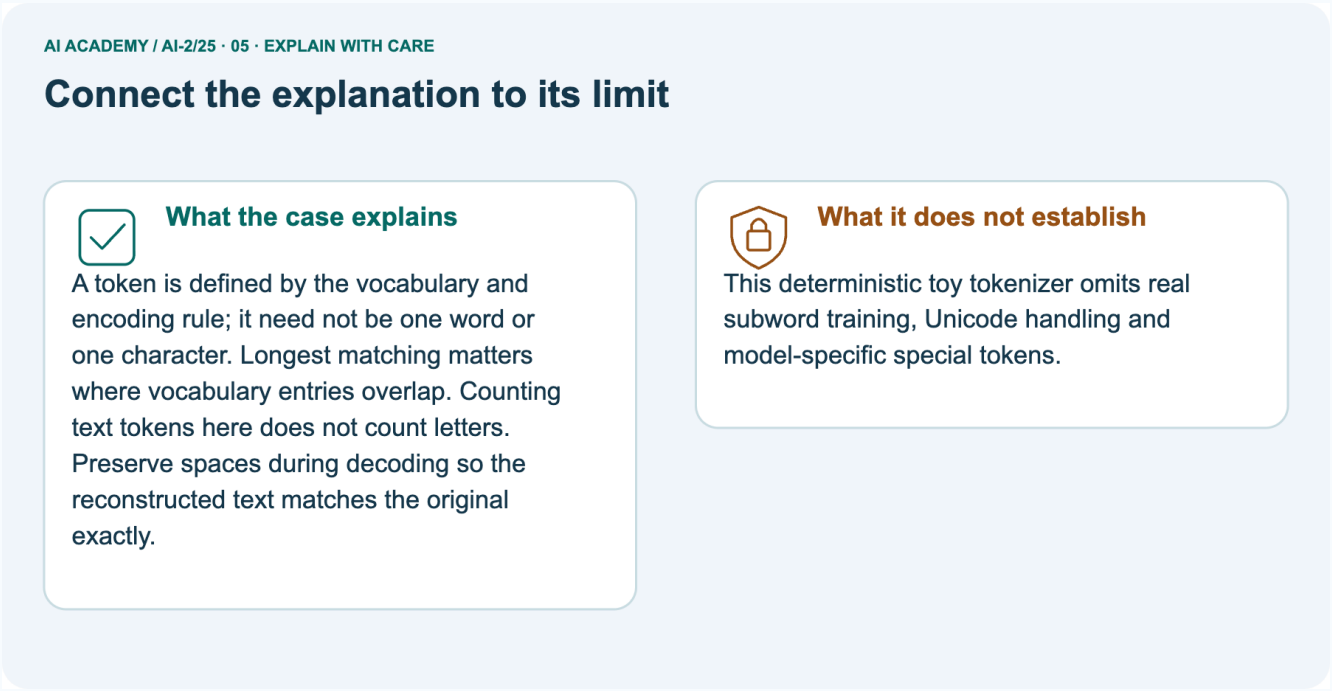
Original case: For books book!, the ! at position 10 has no **token**: stop with error. Changed condition: Use a character **vocabulary** covering b,o,k,s and space instead. Expected result: The same phrase uses ten character **token**s.

Reason: There are five letters, one space and four letters; its decoded text remains unchanged.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

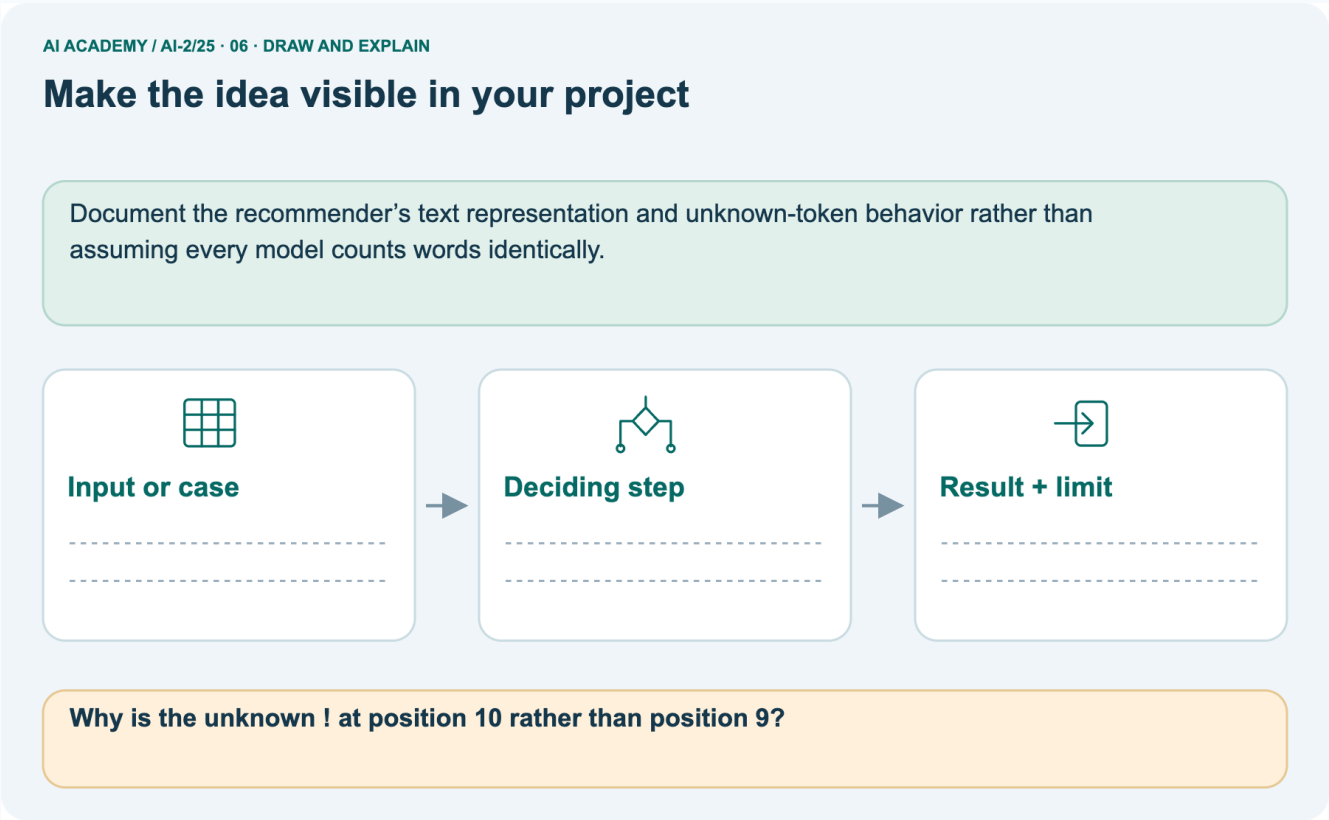
A **token** is defined by the **vocabulary** and **encoding** rule; it need not be one word or one character. Longest matching matters where **vocabulary** entries overlap. Counting text **token**s here does not count letters.

Preserve spaces during **decoding** so the reconstructed text matches the original exactly. Limit: This deterministic toy **token**izer omits real subword **training**, Unicode handling and model-specific special **token**s.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document the recommender's text representation and unknown-**token** behavior rather than assuming every model counts words identically. Draw your **input**, deciding step, result and limit.

Supported question: Why is the unknown ! at position 10 rather than position 9?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Text as Pieces: the case

02. Follow the deciding steps

03. Keep the evidence roles separate

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.